

# Rounding to Named Places

## Answer Key

Grades 4–5 Mathematics

### Quick Answers

---

- |          |                                                  |
|----------|--------------------------------------------------|
| 1. 400   | 7. 50000                                         |
| 2. 4240  | 8. 7400                                          |
| 3. 5670  | 9. nearest ten = 26490, nearest hundred = 26500, |
| 4. 5700  | nearest thousand = 26000                         |
| 5. 6000  | 10. the correct rounded number = 4000, —         |
| 6. 18000 |                                                  |
- 

### What is verified

---

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

---

### Curriculum

---

**Level:** Grades 4–5 Mathematics

4.NBT.A.3 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

*Difficulty 1–4 of 5 across 13 tagged checks.*

---

### Common wrong answers

---

10. *If they got 3900: rounded the hundreds digit down, ignoring that the tens digit 5 pushes 39 hundreds up to 40*

---

- 1.** Round 362 to the nearest hundred (number line 300–400).

**Solution:** The midpoint between 300 and 400 is 350. On the line, 362 sits to the *right* of 350, so it is closer to 400. 400

- 2.** Round 4,238 to the nearest ten (number line 4,230–4,240).

**Solution:** The midpoint is 4,235, and 4,238 sits past it — closer to 4,240. 4240

- 3.** Round 5,671 to the nearest ten.

**Solution:** Underline the tens digit (7); look one place right at the ones digit:  $1 < 5$ , so keep the 7 tens and zero the ones. 5670

4. Round 5,671 to the nearest hundred.

**Solution:** Underline the hundreds digit (6); the tens digit is  $7 \geq 5$ , so round the 6 hundreds up to 7 hundreds. 5700

5. Round 5,671 to the nearest thousand.

**Solution:** Underline the thousands digit (5); the hundreds digit is  $6 \geq 5$ , so round the 5 thousands up to 6 thousands. 6000

6. 18,437 tickets, reported to the nearest thousand.

**Solution:** The thousands digit is 8; the hundreds digit is  $4 < 5$ , so the thousands digit stays and everything to the right becomes zero. The paper prints 18,000 tickets. 18000

7. Round 49,850 to the nearest thousand.

**Solution:** The thousands digit is 9; the hundreds digit is  $8 \geq 5$ , so round up. But 49 thousands rounding up becomes 50 thousands — the carry changes the ten-thousands digit too:  $49,850 \rightarrow 50,000$ . 50000

8. 7,352 dollars, rounded to the nearest hundred dollars.

**Solution:** The hundreds digit is 3; the tens digit is  $5 \geq 5$ , so round the 3 hundreds up to 4 hundreds: 7,400 dollars. 7400

9. Round the population 26,489 to the nearest ten, hundred, and thousand.

**Solution:** Nearest ten: ones digit  $9 \geq 5$ , so 26,490. 26490

Nearest hundred: tens digit  $8 \geq 5$ , so 4 hundreds become 5: 26,500. 26500

Nearest thousand: hundreds digit  $4 < 5$ , so the 6 thousands stay: 26,000. 26000

10. Kai says 3,952 rounds to 4,000 (nearest hundred); Jo says 3,900. Who is correct?

**Solution:** The hundreds digit is 9 and the tens digit is  $5 \geq 5$ , so the 9 hundreds round up: 39 hundreds become 40 hundreds, and the carry gives 4,000. Kai is correct. 4000

Full credit for the explanation: the tens digit 5 means round up; Jo rounded down; and rounding 39 hundreds up to 40 hundreds carries into the thousands place, which is why the answer looks like a thousand.